

E-Commerce Customer Analytics & Churn Analysis

Overview

This project analyzes customer behavior in an e-commerce platform using a dataset containing demographic information, purchasing patterns, loyalty metrics, customer satisfaction scores, and churn indicators.

The objective is to identify key factors affecting customer retention, spending habits, and customer lifetime value while generating actionable business insights.

Business Problem

Customer acquisition is expensive, making customer retention a critical factor for long-term business growth. This project aims to answer:

- Which customers generate the highest revenue?
- What factors influence customer churn?
- How do loyalty programs affect spending behavior?
- Which customer segments are most valuable?
- How does customer satisfaction impact retention?

Dataset Information

Dataset Size

- Records: 1,200 Customers
- Features: 25 Variables

Key Features

Feature	Description
age	Customer Age
gender	Gender
country	Customer Country
region	Region
income_bracket	Income Level
customer_lifetime_days	Customer Tenure
total_orders	Number of Orders
total_spent_usd	Total Spending
avg_order_value_usd	Average Order Value

Feature	Description
preferred_category	Favorite Product Category
loyalty_tier	Loyalty Membership Level
discount_usage_rate	Discount Usage
return_rate	Product Return Rate
satisfaction_score	Customer Satisfaction
churn	Customer Churn Indicator

Project Workflow

1. Data Collection
 2. Data Cleaning
 3. Exploratory Data Analysis (EDA)
 4. Customer Segmentation
 5. Churn Analysis
 6. Business Insights
 7. Recommendations
-

Tools & Technologies

- Python
 - Pandas
 - Jupyter Notebook
 - Power BI
 - MS Excel
-

Key Analyses Performed

Customer Demographics Analysis

- Age distribution
- Gender distribution
- Regional analysis
- Income bracket analysis

Spending Behavior Analysis

- Total spending trends
- Average order value analysis
- High-value customer identification

Loyalty Program Analysis

- Loyalty tier distribution
- Spending by loyalty tier
- Retention by loyalty tier

Customer Satisfaction Analysis

- Satisfaction score distribution
- Relationship between satisfaction and churn

Churn Analysis

- Churn rate calculation
 - Factors affecting churn
 - Customer retention insights
-

Business Insights

- Identified customer groups contributing the highest revenue.
 - Evaluated the impact of loyalty programs on customer retention.
 - Measured the relationship between customer satisfaction and churn.
 - Determined behavioral patterns associated with high-value customers.
-

Recommendations

- Strengthen loyalty programs to improve retention.
 - Target high-value customer segments with personalized marketing.
 - Reduce churn through proactive customer engagement.
 - Improve customer satisfaction through enhanced service quality.
 - Develop retention strategies for at-risk customers.
-

Future Scope

- Machine Learning-based Churn Prediction
 - Customer Segmentation using Clustering
 - Customer Lifetime Value Prediction
-

Author

Anuj Singh

GitHub: <https://github.com/Devil-shree>